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IOM205 Advanced Database Management

Individual Coursework Report

UniBooks Database

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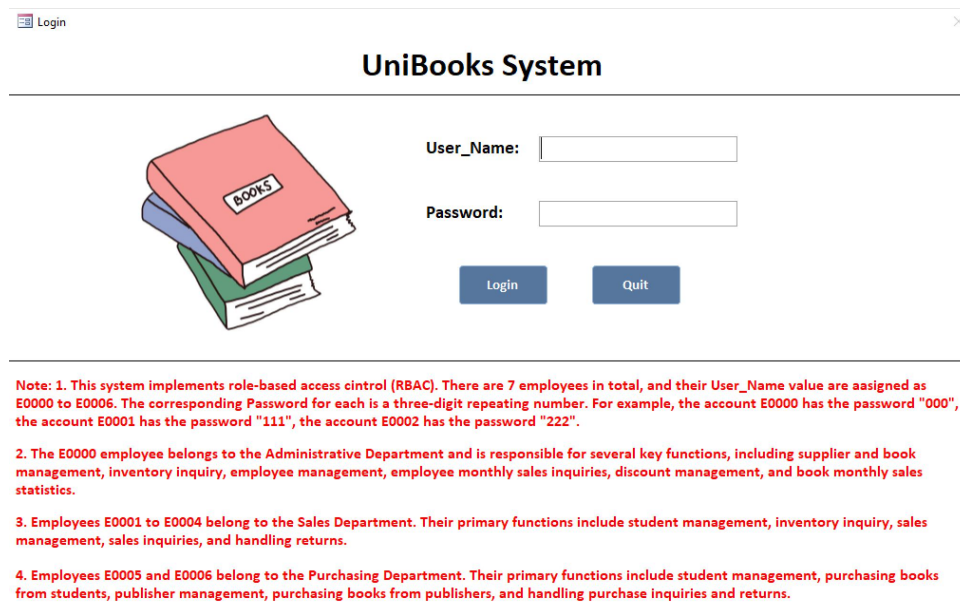
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ID2.2: The operation of the system

2.2.1 Log in

Double-click to open Access Database, a login interface will appear, users need to enter the appropriate account and password to log in, as shown in the following figure:



The login interface for the UniBooks System. It features a window titled "Login" with a close button. The main title is "UniBooks System". On the left is an illustration of a stack of books. To the right are input fields for "User_Name:" and "Password:". Below these are "Login" and "Quit" buttons. At the bottom, there is a red note containing system details and employee roles.

User_Name:

Password:

Login Quit

Note: 1. This system implements role-based access control (RBAC). There are 7 employees in total, and their User_Name value are assigned as E0000 to E0006. The corresponding Password for each is a three-digit repeating number. For example, the account E0000 has the password "000", the account E0001 has the password "111", the account E0002 has the password "222".

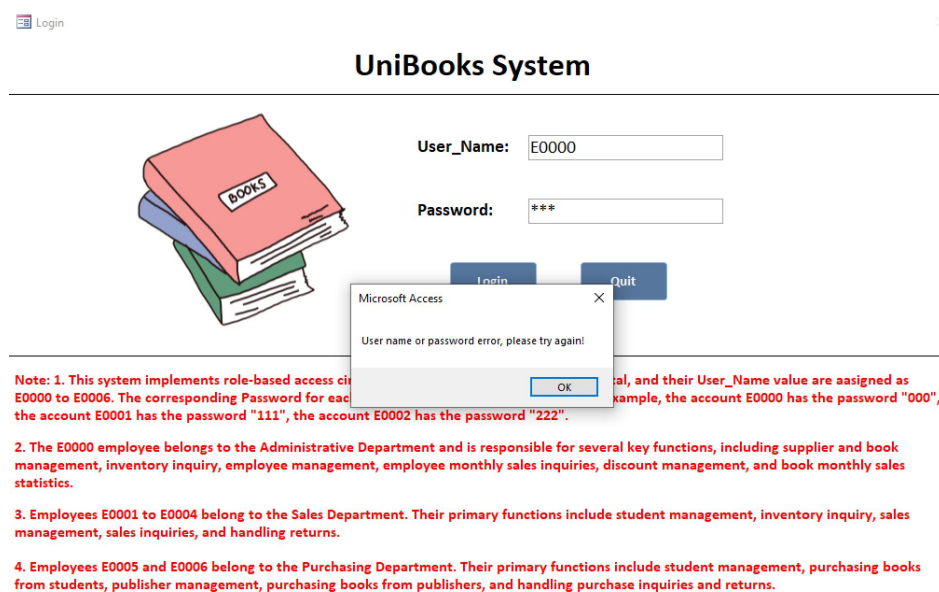
2. The E0000 employee belongs to the Administrative Department and is responsible for several key functions, including supplier and book management, inventory inquiry, employee management, employee monthly sales inquiries, discount management, and book monthly sales statistics.

3. Employees E0001 to E0004 belong to the Sales Department. Their primary functions include student management, inventory inquiry, sales management, sales inquiries, and handling returns.

4. Employees E0005 and E0006 belong to the Purchasing Department. Their primary functions include student management, purchasing books from students, publisher management, purchasing books from publishers, and handling purchase inquiries and returns.

Figure1: UniBooks Login Interface

However, if the account number and password are entered incorrectly, an error screen will pop up, as shown in the figure:



The login interface for the UniBooks System, showing an error message. The "User_Name:" field contains "E0000" and the "Password:" field contains "***". A "Microsoft Access" error dialog box is displayed in the foreground, stating "User name or password error, please try again!" with an "OK" button. The background interface is identical to Figure 1, including the book illustration, buttons, and red note.

User_Name: E0000

Password: ***

Login Quit

Microsoft Access

User name or password error, please try again!

OK

Note: 1. This system implements role-based access control (RBAC). There are 7 employees in total, and their User_Name value are assigned as E0000 to E0006. The corresponding Password for each is a three-digit repeating number. For example, the account E0000 has the password "000", the account E0001 has the password "111", the account E0002 has the password "222".

2. The E0000 employee belongs to the Administrative Department and is responsible for several key functions, including supplier and book management, inventory inquiry, employee management, employee monthly sales inquiries, discount management, and book monthly sales statistics.

3. Employees E0001 to E0004 belong to the Sales Department. Their primary functions include student management, inventory inquiry, sales management, sales inquiries, and handling returns.

4. Employees E0005 and E0006 belong to the Purchasing Department. Their primary functions include student management, purchasing books from students, publisher management, purchasing books from publishers, and handling purchase inquiries and returns.

Figure2: Account or password error screen

Annotation: The passwords for the respective accounts are as follows:

- E0000: 000
- E0001: 111
- E0002: 222
- E0003: 333
- E0004: 444
- E0005: 555
- E0006: 666

This system adopts a **decentralized management approach**, dividing UniBooks employees into three departments: The **Administrative Department** manages employees, discounts, and operational data; the **Sales Department** handles sales and returns; and the **Purchasing Department** oversees procurement and related returns. The structure is shown in the diagram below:

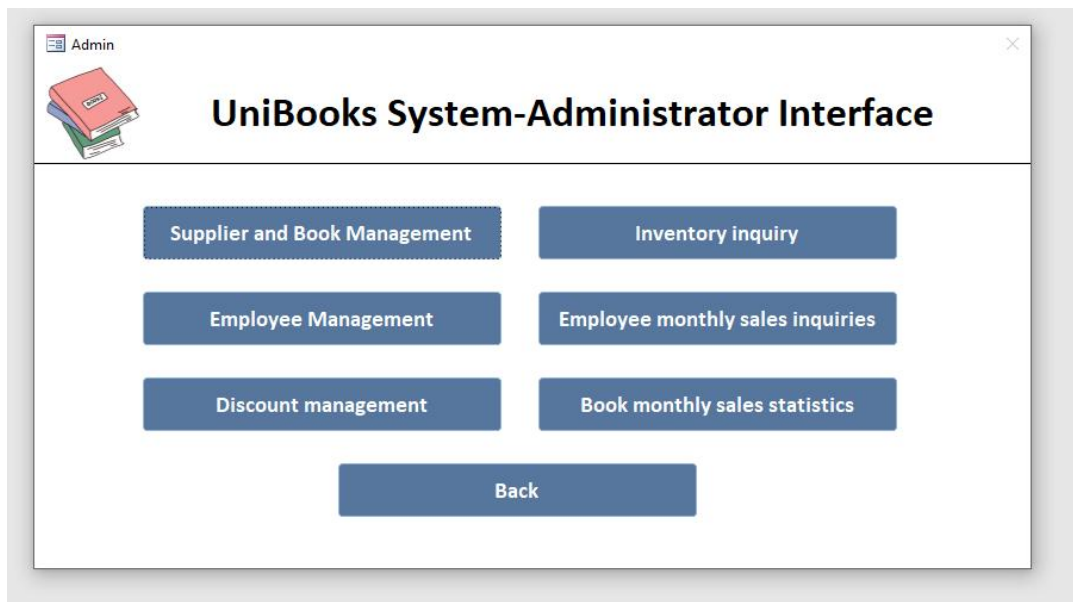


Figure3: UniBooks System-Administrator Interface

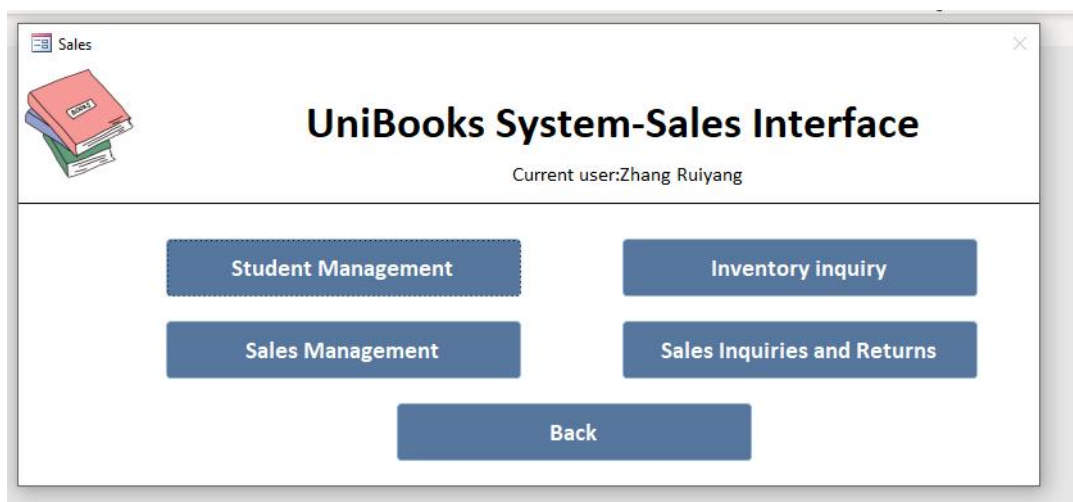


Figure4: UniBooks System-Sales Interface

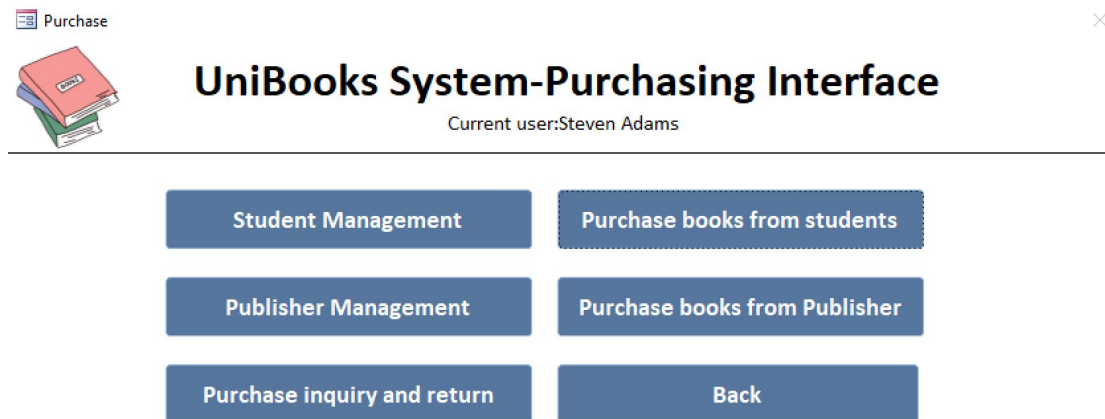


Figure5: UniBooks System-Purchasing Interface

2.2.2 Specific Operations and Usage

Administrative Department employees can log in to perform strategic tasks like Employee, Supplier, Book, and Discount Management, and view detailed sales data, as shown in the diagram below.

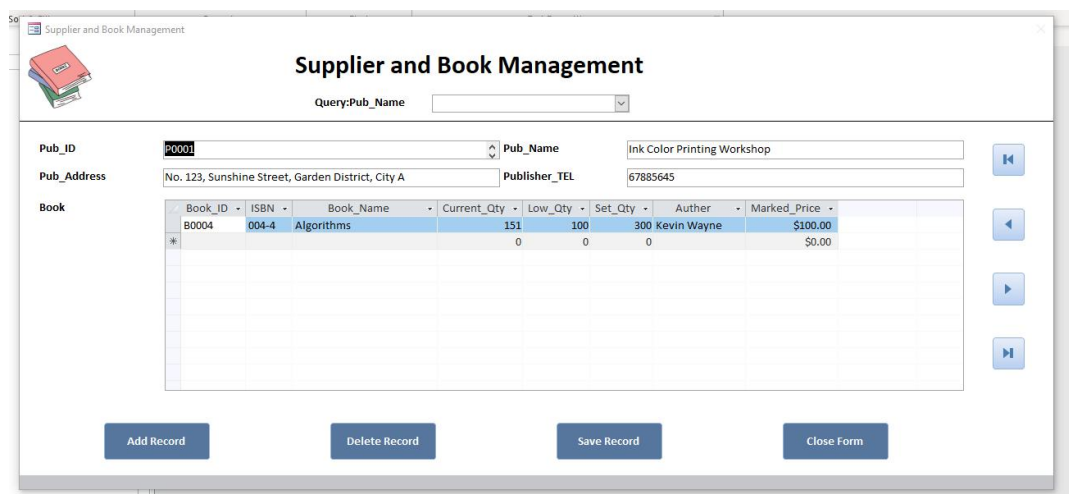
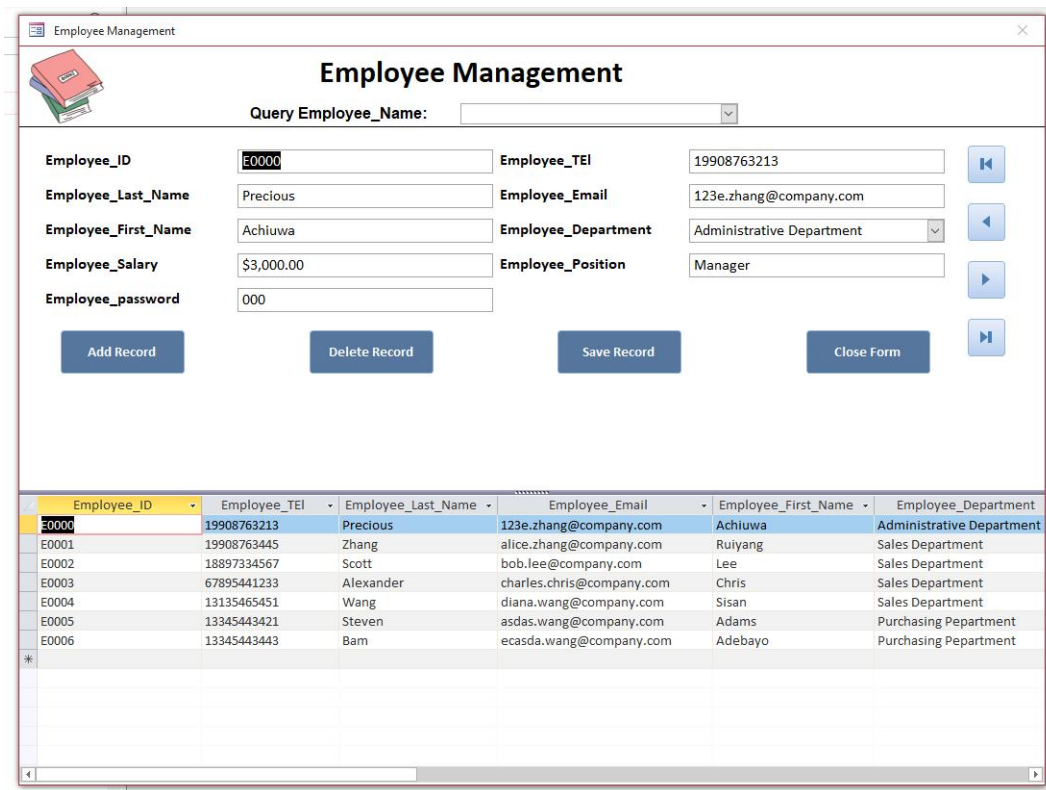


Figure6: Supplier and Book Management

Annotation: Here, you can not only manage publisher information but also add books published by the corresponding publisher. Additionally, you can use the query function above to quickly search for information about their published books.



Employee Management

Query Employee_Name:

Employee_ID: Employee_TEI:

Employee_Last_Name: Employee_Email:

Employee_First_Name: Employee_Department:

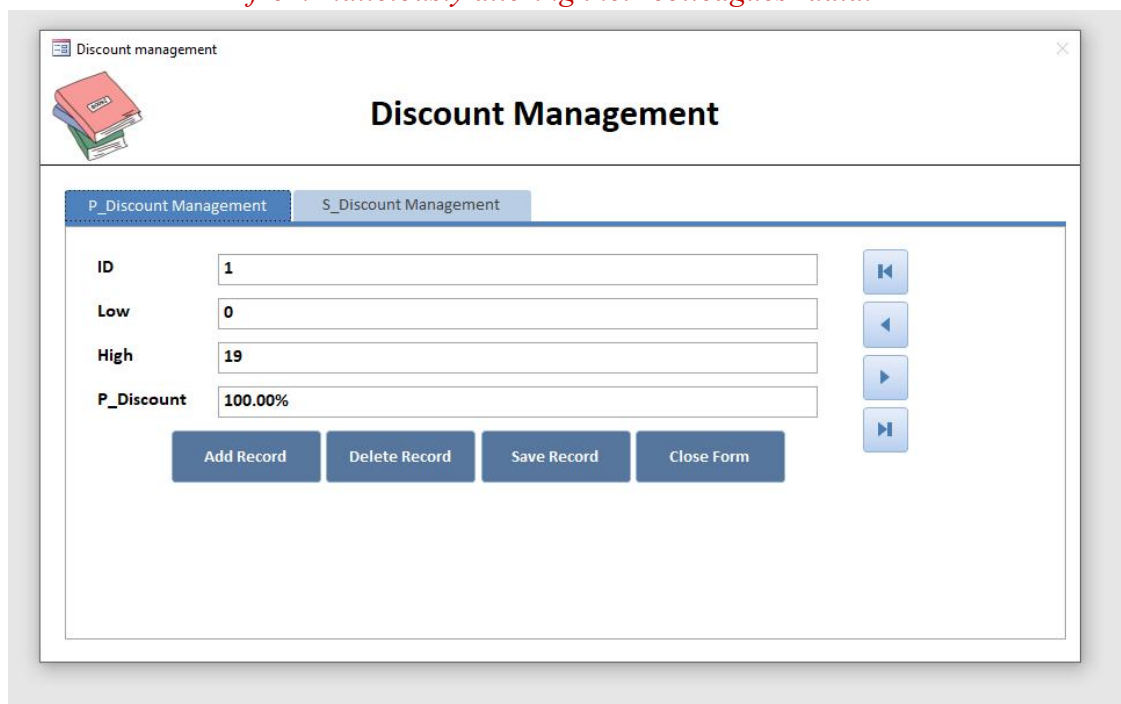
Employee_Salary: Employee_Position:

Employee_password:

Employee_ID	Employee_TEI	Employee_Last_Name	Employee_Email	Employee_First_Name	Employee_Department
E0000	19908763213	Precious	123e.zhang@company.com	Achiuwa	Administrative Department
E0001	19908763445	Zhang	alice.zhang@company.com	Ruiyang	Sales Department
E0002	18897334567	Scott	bob.lee@company.com	Lee	Sales Department
E0003	67895441233	Alexander	charles.chris@company.com	Chris	Sales Department
E0004	13135465451	Wang	diana.wang@company.com	Sisan	Sales Department
E0005	13345443421	Steven	asd.a.wang@company.com	Adams	Purchasing Department
E0006	13345443443	Bam	ecad.a.wang@company.com	Adebayo	Purchasing Department

Figure7: Employee Management

Annotation: Due to the **decentralized management system**, only system administrators are authorized to modify employee information, preventing employees from maliciously altering their colleagues' data.



Discount Management

P_Discount Management S_Discount Management

ID:

Low:

High:

P_Discount:

Figure8: Discount Management

Annotation: In Discount Management, new discounts can be added. For example, sellers may receive an 80% discount if they sell more than 80 books in the future.

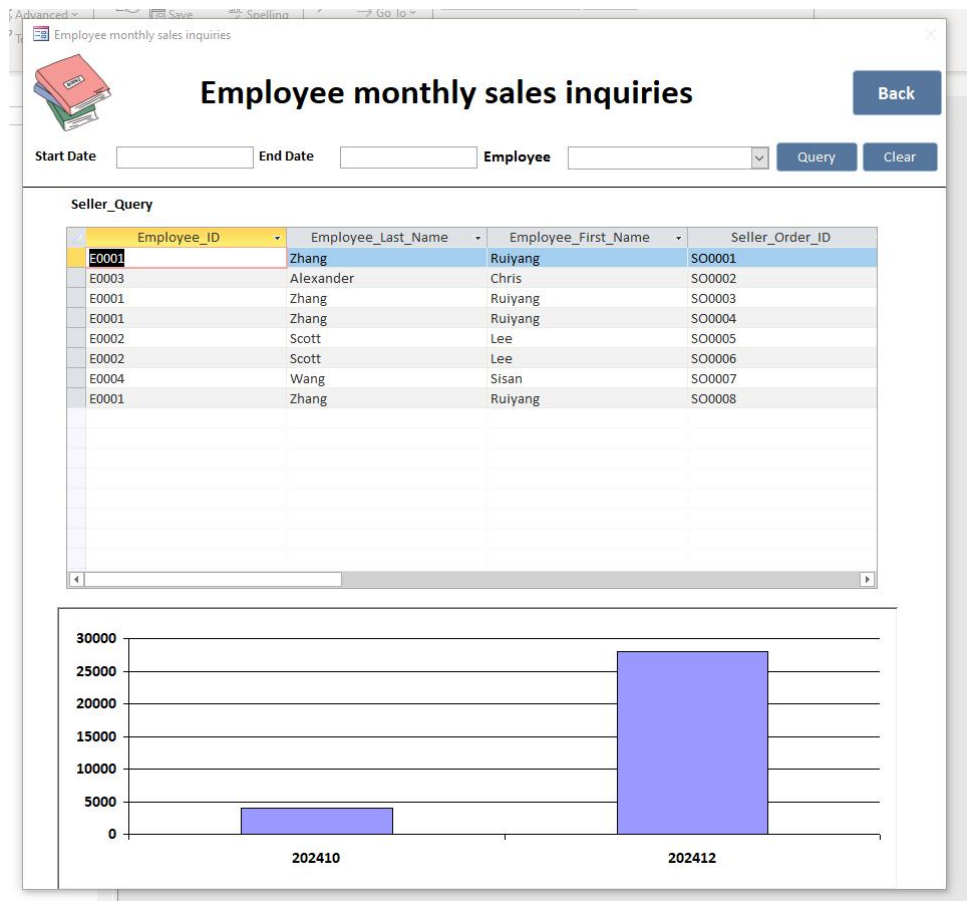


Figure9: Employee monthly sales inquiries

Annotation: The specific details will be elaborated in the chart section later.

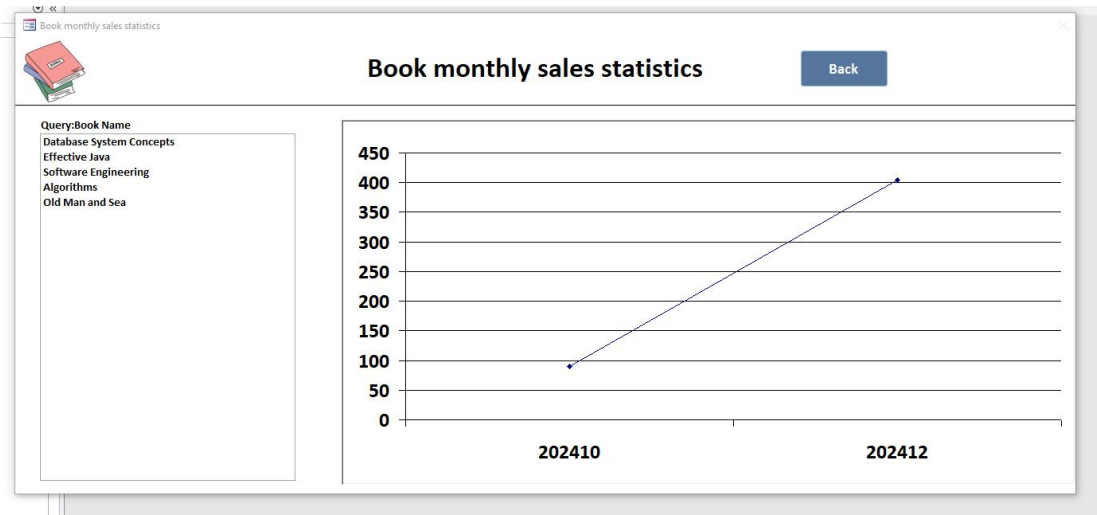


Figure10: Book monthly sales statistics

Annotation: The specific details will be elaborated in the chart section later.

If the employee belongs to the **Sales Department**, they can perform functions such as Inventory Inquiry, Sales Management, Sales Inquiries, and Returns.

Student Management

Query Last_Name:

Stu_ID:

Last_Name:

First_Name:

Student_TEL:

Student_Email:

Address:

Buttons: Add Record, Delete Record, Save Record, Close Form

Figure11: Student Management

*Annotation: You can use queries to quickly search for student information.
At this point, the student is the target customer for our book sales.*

Sales Management

Order_ID: Create

Student:

Order_Date:

Employee:

Order_Qty:

S_DiscountID:

Order_Price:

TrackingNumber:

Book	Book_Qty	Price
Algorithms	90	\$80.00
Software Engineering	80	\$48.00
Old Man and Sea	20	\$128.00
*	1	

Buttons: Add, Back

Figure12: Sales Management

*Annotation: In this form, the Price in the subform is 80% of the book's marked price.
The Order_Qty automatically sums up based on Book_Qty, while S_Discount and Order_Price are auto-calculated. The Employee field auto-fills the current account to distinguish orders, and Order_Date is fixed to the current day, preventing edits to past or future dates.*

Inventory inquiry

ISBN Book_Name Pub_Name

Book_ID	ISBN	Book_Name	Current_Qty	Low_Qty	Set_Qty	Author	Marked_Price	Pub_Name
B0004	004-4	Algorithms	152	100	300	Kevin Wayne	\$100.00	Ink Color Printing Workshop
B0005	005-5	Old Man and Sea	80	60	40	Ernest Miller Hemingw	\$160.00	Exquisite Art Printing
B0001	001-1	Database System Concepts	51	80	40	S.Sudarshan	\$20.00	Brilliant Printing Factory
B0002	002-2	Effective Java	141	50	50	Joshua Blosh	\$30.00	Brilliant Printing Factory
B0003	003-3	Software Engineering	126	80	100	Ian Sommerville	\$60.00	Creativity Beauty Printing Bureau

Figure13: Inventory inquiry

Annotation: When Current_Qty is less than Low_Qty, a red marker will automatically warn the user.

Sales Query

Sales Inquiries and Returns

Current user:Zhang Ruiyang

Start Date End Date Book Name

Sales_Query

Order_ID	Stu_Name	Order_Date	P_DiscountID	Book_ID	ISBN	Book_Name	Book_Qty	Return or not	
SO0001	Liu Yida	10/4/2024		B0003	003-3	Software Engineering	10	☒	Book Damage
SO0003	Zhang Xinyuan	10/12/2024		B0003	003-3	Software Engineering	20	☐	
SO0003	Zhang Xinyuan	10/12/2024		B0004	004-4	Algorithms	30	☐	
SO0004	Zhang Xinyuan	12/16/2024	87.50%	B0004	004-4	Algorithms	60	☐	
SO0008	Lin Nan	12/16/2024	90.00%	B0005	005-5	Old Man and Sea	30	☐	

Note: Double-click the order ID to open the corresponding report

Figure14: Sales Inquiries and Returns

Annotation: Double-click the **Order_ID** can open the corresponding report
The specific return operation will be demonstrated in detail below.

Purchase books from students

Student_Buyer_Order_ID: SBO0006 Create Buyer_Order_Price: \$272.00

Student: Su Shenghe Employee: Steven Adams

Buyer_Order_Date: 12/17/2024 TrackingNumber: FSADFAIWBCHDG

Buyer_Order_Qty: 18

Book	Book_Qty	Price
Algorithms	9	\$20.00
Database System Concepts	7	\$4.00
Old Man and Sea	2	\$32.00
*	1	

Add Back

Figure17: Purchase books from students

Annotation: This process is similar to the sales process, where the book quantity and price are automatically calculated. The only difference is that purchasing books from students does not include any discounts.

Publisher Management

Query Pub_Name:

Pub_ID: P0001

Pub_Name: Ink Color Printing Workshop

Pub_Address: No. 123, Sunshine Street, Garden District, City A

Publisher_TEL: 67885645

Add Record Delete Record Save Record Close Form

Figure18: Pulisher Management

Annotation: You can use queries to quickly search for Publisher information.

Purchase books from Publisher

[illegible]

Figure19: Purchase books from Publisher

Annotation: The specific operations are similar to the sales process.


Purchase inquiry and return



Purchase inquiry and return

Current user:Steven Adams

Back

Date Start
End Start
Book Name

Query

Clear

Publisher_Buyer_Query

Order_ID	Pub_Name	Order_Date	P_DiscountID	Book_ID	ISBN	Book_Name	Book_Qty	Return or not
BO0005	Ink Color Printing Workshop	12/15/2024	100.00%	B0001	001-1	Database System Concepts	20	☐
BO0005	Ink Color Printing Workshop	12/15/2024	100.00%	B0002	002-2	Effective Java	1	☐
BO0005	Ink Color Printing Workshop	12/15/2024	100.00%	B0003	003-3	Software Engineering	1	☐
BO0006	Ink Color Printing Workshop	12/15/2024	95.00%	B0001	001-1	Database System Concepts	20	☐
BO0006	Ink Color Printing Workshop	12/15/2024	95.00%	B0002	002-2	Effective Java	5	☐
BO0006	Ink Color Printing Workshop	12/15/2024	95.00%	B0003	003-3	Software Engineering	5	☐
BO0007	Exquisite Art Printing	12/16/2024	95.00%	B0005	005-5	Old Man and Sea	40	☐
BO0009	Brilliant Printing Factory	12/17/2024	95.00%	B0001	001-1	Database System Concepts	30	☐

Student_Buyer_Query

Order_ID	Stu_Name	Order_Date	Book_ID	ISBN	Book_Name	Book_Qty	Return or not	Return
SBO0005	Su Shenghe	12/15/2024	B0003	003-3	Software Engineering	1	☒	Book Damage
SBO0005	Su Shenghe	12/15/2024	B0004	004-4	Algorithms	1	☐	
SBO0005	Su Shenghe	12/15/2024	B0001	001-1	Database System Concepts	1	☐	

Note: Double-click the order ID to open the corresponding report

Figure20: Purchase inquiry and return

Annotation: The specific operations are similar to the aforementioned Sales Inquiries and Returns.

ID2.3: Significant Database Features

The operations for sales and purchases are the same; the following examples will use sales as an illustration.

2.3.1 Adds a new order and updates stock based on order size

This functionality combines order creation and inventory updates into **one step**, reducing workload, minimizing errors from manual updates, and allowing employees to focus on more critical tasks, thereby improving efficiency (Chiru, 2015).

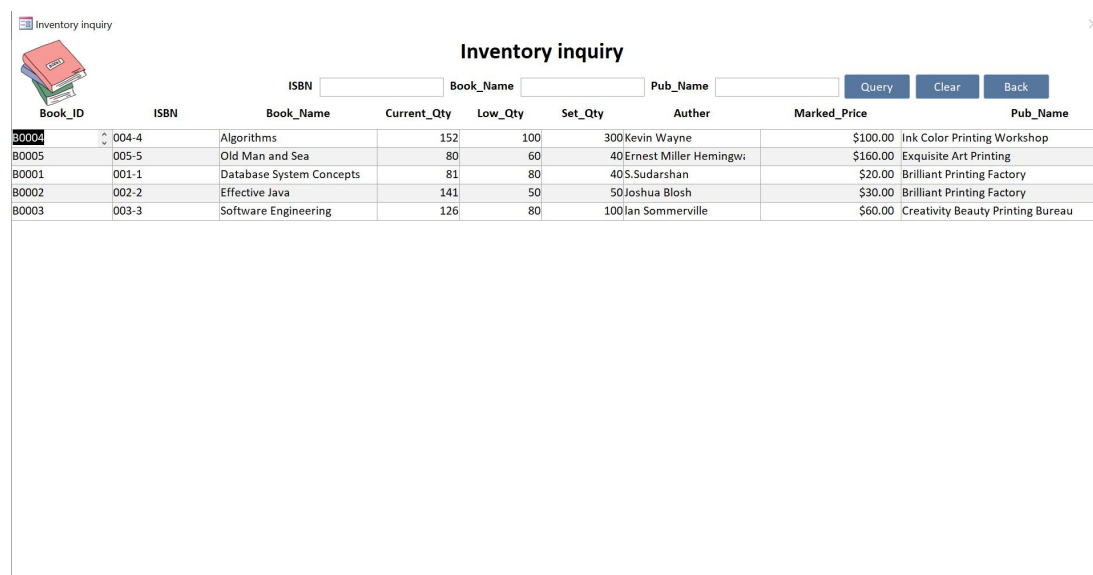
Additionally, due to the system's **decentralized management mechanism**, orders are automatically assigned to the current account's employee. This ensures clear accountability and prevents order mix-ups, as each order is distinctly linked to the responsible employee. Such a mechanism enhances data accuracy, simplifies tracking, and promotes a well-organized workflow, which is essential for maintaining smooth operations within the company.

Annotation: Notably, during this step:

- 1. The total book quantity (**Order_Qty**) is automatically calculated by summing up the detailed **Book_Qty** from the subform.*
- 2. **S_Discount** is determined based on the total book quantity.*
- 3. **Order_Price** is also automatically calculated.*

This automation eliminates the need for manual calculations, greatly improving work efficiency.

The process of creating an order and updating inventory in a **single step** is as follows: First, this is the inventory before the order is created. As shown, *Old Man and Sea* and *Database System Concepts* have **80** and **81** copies, respectively.



Book_ID	ISBN	Book_Name	Current_Qty	Low_Qty	Set_Qty	Author	Marked_Price	Pub_Name
B0004	004-4	Algorithms	152	100	300	Kevin Wayne	\$100.00	Ink Color Printing Workshop
B0005	005-5	Old Man and Sea	80	60	40	Ernest Miller Hemingw	\$160.00	Exquisite Art Printing
B0001	001-1	Database System Concepts	81	80	40	S.Sudarshan	\$20.00	Brilliant Printing Factory
B0002	002-2	Effective Java	141	50	50	Joshua Blosh	\$30.00	Brilliant Printing Factory
B0003	003-3	Software Engineering	126	80	100	Ian Sommerville	\$60.00	Creativity Beauty Printing Bureau

Figure21: The inventory before any operation has been performed

Next, we purchase **30 copies** of each of these two books.

Sales Management

Order_ID

S00009

▼

Create

Student

S0001

▼

Order_Date

12/17/2024

Employee

Zhang Ruiyang

▼

Order_Qty

60

S_DiscountID

87.50%

▼

Order_Price

\$3,780.00

TrackingNumber

GUFREKBKBTBH

Seller_Order_Item

Book	Book_Qty	Price
Database System Concepts	30	\$16.00
Old Man and Sea	30	\$128.00
*	1	

Add

Back

Figure22: Order Placement Interface

Sales Management

Order_ID

S00009

▼

Create

Student

S0001

▼

Order_Date

12/17/2024

Employee

Zhang Ruiyang

▼

Order_Qty

60

S_DiscountID

87.50%

▼

Order_Price

\$3,780.00

TrackingNumber

GUFREKBKBTBH

Seller_Order_Item

Book	Book_Qty	Price
Database System Concepts	30	\$16.00
Old Man and Sea	30	\$128.00
*	1	

Add

Back

Microsoft Access

Add successfully!

OK

Figure23: Order successfully placed

Then, checking the inventory levels reveals that the **Current_Qty** for both books has decreased by **30 copies**. Since both quantities are now below the **Low_Qty**, they are marked in red.

Inventory inquiry

Book_ID	ISBN	Book_Name	Current_Qty	Low_Qty	Set_Qty	Author	Marked_Price	Pub_Name
B0004	004-4	Algorithms	152	100	300	Kevin Wayne	\$100.00	Ink Color Printing Workshop
B0005	005-5	Old Man and Sea	50	60	40	Ernest Miller Hemingwa	\$160.00	Exquisite Art Printing
B0001	001-1	Database System Concepts	51	80	40	S.Sudarshan	\$20.00	Brilliant Printing Factory
B0002	002-2	Effective Java	141	50	50	Joshua Blosh	\$30.00	Brilliant Printing Factory
B0003	003-3	Software Engineering	126	80	100	Ian Sommerville	\$60.00	Creativity Beauty Printing Bureau

Figure24: Inventory levels after placing the order

Finally, my employee's sales records are also updated simultaneously.

Sales Query

Sales Inquiries and Returns

Current user:Zhang Ruiyang

Start Date End Date Book Name Query Clear Back

Order_ID	Stu_Name	Order_Date	P_DiscountID	Book_ID	ISBN	Book_Name	Book_Qty	Return or not	
SO0009	Liu Yida	12/17/2024	87.50%	B0001	001-1	Database System Concepts	30	☐	
SO0009	Liu Yida	12/17/2024	87.50%	B0005	005-5	Old Man and Sea	30	☐	
SO0001	Liu Yida	10/4/2024		B0003	003-3	Software Engineering	10	☒	Book Damage
SO0003	Zhang Xinyuan	10/12/2024		B0003	003-3	Software Engineering	20	☐	
SO0003	Zhang Xinyuan	10/12/2024		B0004	004-4	Algorithms	30	☐	
SO0004	Zhang Xinyuan	12/16/2024	87.50%	B0004	004-4	Algorithms	60	☐	
SO0008	Lin Nan	12/16/2024	90.00%	B0005	005-5	Old Man and Sea	30	☒	Book Pring Err

Note: Double-click the order ID to open the corresponding report

Figure25: Employee records are updated simultaneously

2.3.2 Return Processing with One-Step Inventory Update

The system efficiently handles real-world return scenarios, including partial and multiple returns for a single order. This streamlines operations, reduces manual workload, minimizes errors, and enhances customer satisfaction by offering a smoother return process. The specific operations are as follows:

First, open **Sales Inquiries and Returns**, select the details of the recently placed order, check the **Return or Not** attribute, and enter the return reason.

ID2.4: Non-Trivial queries

In this section, I will use **three non-trivial queries** to explain in detail how I create orders and update data.

2.4.1 Step1: Initial State - No Data Binding Established

Initially, there is no data binding between the **Seller_Order** main form and the **Seller_Order_Item** subform. The main form shows no data, the subform is empty, and users can only manually add order details in the subform.

2.4.2 Step2: Entering Data in the Seller_Order_Item Subform

At this stage, the user manually enters the order details into the **Seller_Order_Item** subform. However, the **Seller_Order_ID** in the subform is intentionally left blank, resulting in a NULL value. The data source of the subform is based on the following SQL query:

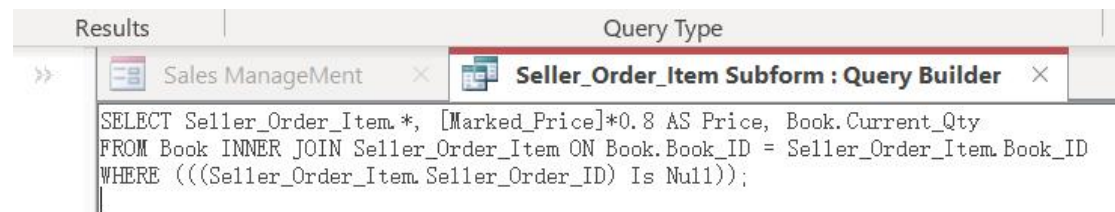


Figure28: Seller_Order_Item_Query

And its Design View is below:

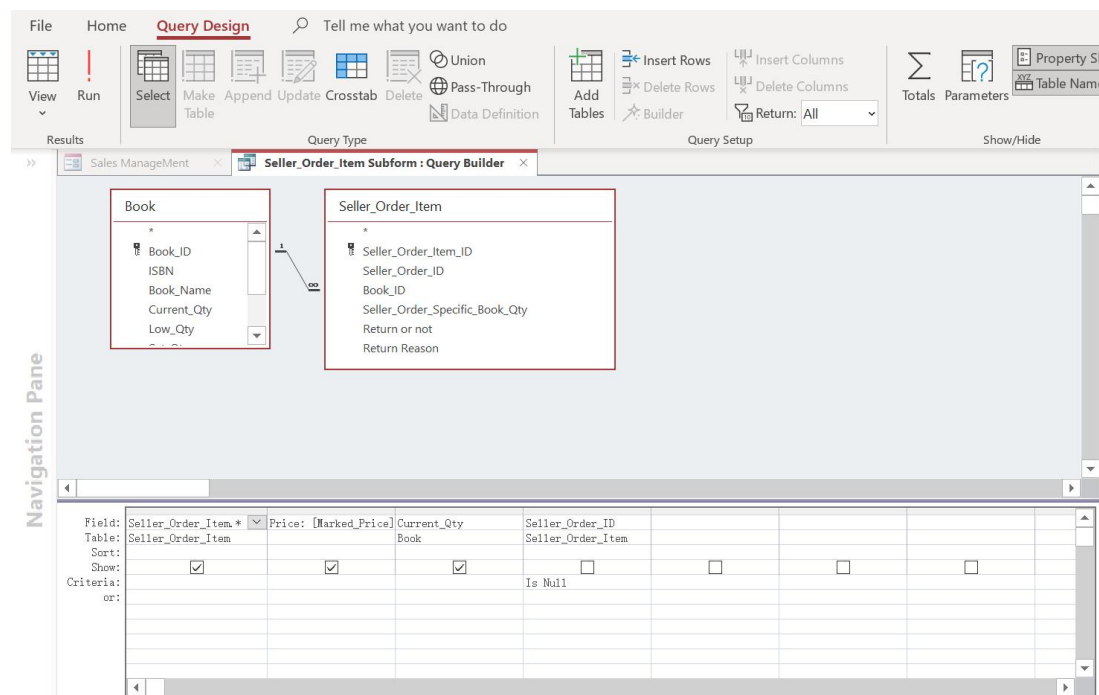


Figure29: Seller_Order_Item_Query Design View

This query retrieves all fields from the **Seller_Order_Item** table, automatically calculates a discounted price (**Price**) by multiplying **Marked_Price** by 0.8, and displays the current stock quantity (**Current_Qty**) from the **Book** table. An INNER JOIN connects the Book table and the **Seller_Order_Item** table through the **Book_ID** field, ensuring that only records with matching **Book_ID** values are included. The condition WHERE **Seller_Order_Item. Seller_Order_ID** Is Null filters the results to show only unassigned order details where **Seller_Order_ID** remains NULL, indicating that these records are not yet associated with any order. In the subform, users can select a book (**Book_ID**) and enter the quantity (**Seller_Order_Specific_Book_Qty**), while the Price field is automatically calculated and cannot be manually edited. The entered data is saved directly into the **Seller_Order_Item** table with **Seller_Order_ID** still being NULL at this stage. The subform dynamically displays these unassigned records, enriched with the book's current stock and discounted price, in preparation for linking them to a specific order in the subsequent steps (Ulrich-Fuller, 2022).

*Annotation: The SELECT statement, which is an inline query, is in the Record Source of the **Seller_Order_Item** subform, which is a subform of the Sales Management main form.*

2.4.3 Step 3: Creating a New Order in the Seller_Order Table

In the main form, users input order details to create a new order. The following INSERT INTO statement is executed to save this data into the **Seller_Order** table:

```
INSERT INTO Seller_Order ( Seller_Order_ID, Stu_ID, Seller_Order_Date,
All_Seller_Order_Qty, All_Seller_Order_Price, Employee_ID, S_DiscountID,
TrackingNumber )
SELECT [forms]![Sales Management]![Seller_Order_ID] AS Expr1,
[forms]![Sales Management]![Stu_ID] AS Expr2,
[forms]![Sales Management]![Seller_Order_Date] AS Expr3,
[forms]![Sales Management]![All_Seller_Order_Qty] AS Expr4,
[forms]![Sales Management]![All_Seller_Order_Price] AS Expr5,
[forms]![Sales Management]![Employee_ID] AS Expr6,
[forms]![Sales Management]![S_DiscountID] AS Expr7,
[forms]![Sales Management]![TrackingNumber] AS Expr8;
```

The Design View is below:

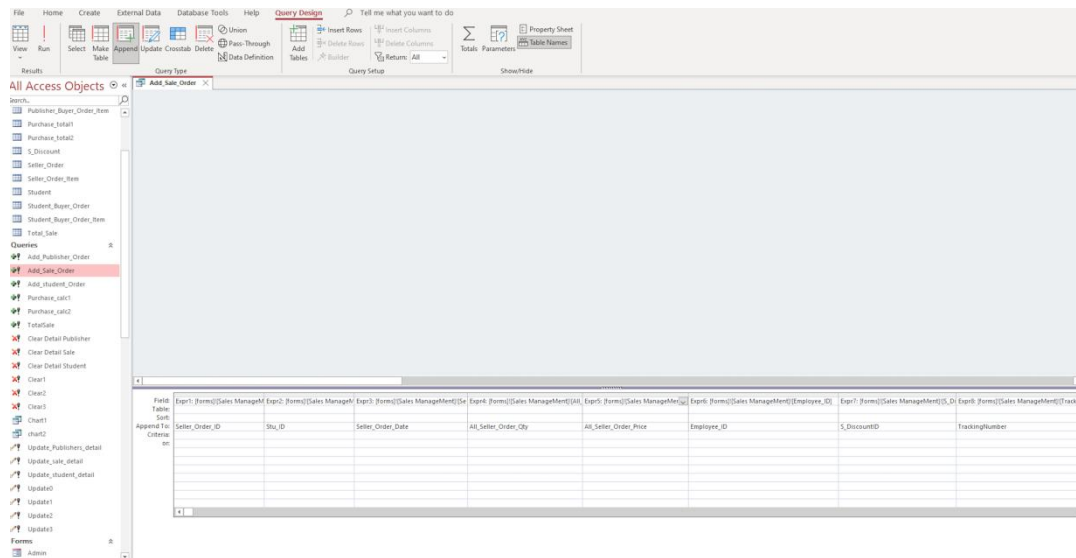


Figure30: Add_Sale_Order Design View

This SQL statement inserts order information from the main form into the **Seller_Order** table. The process is as follows: First, the INSERT INTO clause specifies the target table **Seller_Order** and the fields to be populated: **Seller_Order_ID**, **Stu_ID**, **Seller_Order_Date**, **All_Seller_Order_Qty**, **All_Seller_Order_Price**, **Employee_ID**, **S_DiscountID**, and **TrackingNumber**. Then, the SELECT clause retrieves the corresponding data from the controls on the main form [**Sales Management**] to be inserted into the table. For example, [forms]![Sales Management]![Seller_Order_ID] provides the new order ID, and [forms]![Sales Management]![Stu_ID] provides the student ID. Similarly, other controls' data are mapped to their respective fields. As a result, this SQL statement saves the order information entered in the main form into the **Seller_Order** table, creating a new order record. Each field is populated with the data entered in the main form controls to ensure accuracy. In summary, this SQL statement retrieves user input from the main form and inserts it into the **Seller_Order** table, completing the creation of a new order.

2.4.4 Step 4: Linking Order Details in the Subform to the New Order

The following UPDATE statement is executed:

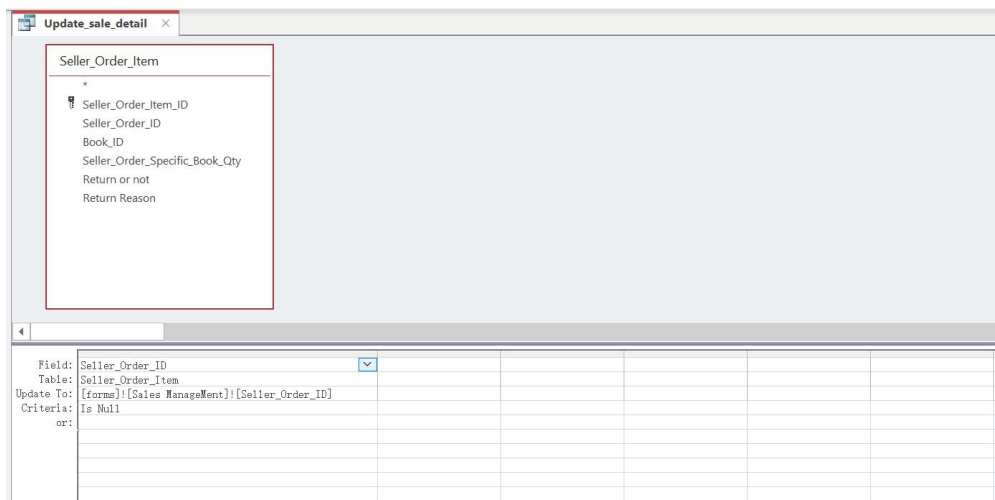


Figure31: Update_sale_detail Design View

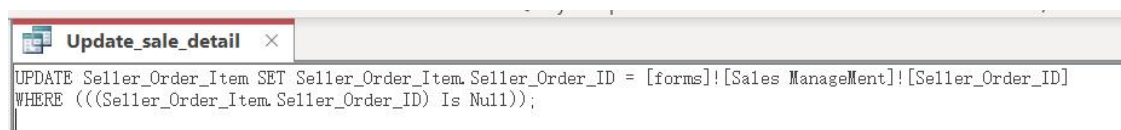


Figure32: Update_sale_detail SQL

This statement updates all order details in the **Seller_Order_Item** table where **Seller_Order_ID** is NULL. The SET clause assigns the new **Seller_Order_ID** from the main form to these unallocated records. As a result, previously unassigned order details are now associated with the new order, This means that the previously intentionally left blank **Seller_Order_ID** in the order details now adopts the **Seller_Order_ID** of the current **Seller_Order**, linking the order details to the main order. This process effectively updates both tables simultaneously (Jiang and Naumann, 2020).

2.4.5 Step 5: Establishing the Data Binding Relationship

At this stage, the main form and subform are linked. Employees fill in order details (excluding **Seller_Order_ID**) in the subform and complete the new order in the main form, synchronized with the **Seller_Order** table. An update query then assigns **Seller_Order_ID** to **Seller_Order_Item**, fully establishing the relationship.

The input and output screenshots are shown below, respectively:



Sales Management

Order_ID **Create** **Order_Qty**
Student **S_DiscountID**
Order_Date **Order_Price**
Employee **TrackingNumber**
Seller_Order_Item

Book	Book_Qty	Price
Database System Concepts	30	\$16.00
Old Man and Sea	30	\$128.00
*	1	

Figure33: Input Screenshot



Sales Inquiries and Returns

Current user:Zhang Ruiyang

Start Date **End Date** **Book Name**

Sales_Query

Order_ID	Stu_Name	Order_Date	P_DiscountID	Book_ID	ISBN	Book_Name	Book_Qty	Return or not	
SO0009	Liu Yida	12/17/2024	87.50%	B0001	001-1	Database System Concepts	30	☐	
SO0009	Liu Yida	12/17/2024	87.50%	B0005	005-5	Old Man and Sea	30	☐	
SO0001	Liu Yida	10/4/2024		B0003	003-3	Software Engineering	10	☒	Book Damage
SO0003	Zhang Xinyuan	10/12/2024		B0003	003-3	Software Engineering	20	☐	
SO0003	Zhang Xinyuan	10/12/2024		B0004	004-4	Algorithms	30	☐	
SO0004	Zhang Xinyuan	12/16/2024	87.50%	B0004	004-4	Algorithms	60	☐	
SO0008	Lin Nan	12/16/2024	90.00%	B0005	005-5	Old Man and Sea	30	☒	Book Pring Err

Note: Double-click the order ID to open the corresponding report

Figure34: Output Screenshot1

Seller_Order_ID	Stu_ID	Seller_Order_Date	All_Seller_Order_Qty	All_Seller_Order_Price	Employee_ID	S_DiscountID	TrackingNumber	Click to Add
SO0001	S0001	10/4/2024	10	\$466.00	E0001		ZX123456789CN	
SO0002	S0002	10/8/2024	30	\$586.00	E0003		MN987654321CN	
SO0003	S0003	10/12/2024	50	\$2,950.00	E0001		PQ456789123CN	
SO0004	S0003	12/16/2024	60	\$4,725.00	E0001	87.50%	DBUAIGFJSBOCV	
SO0005	S0001	12/16/2024	80	\$3,071.25	E0002	87.50%	DBQWUCDSKBREG	
SO0006	S0001	12/16/2024	124	\$8,308.13	E0002	87.50%	HGUFAUBKDBVUG	
SO0007	S0002	12/16/2024	110	\$8,032.50	E0004	87.50%	FWAGDFSBTNGNB	
SO0008	S0004	12/16/2024	30	\$3,888.00	E0001	90.00%	HUIEBWFWUIGGT	
SO0009	S0001	12/17/2024	60	\$3,780.00	E0001	87.50%	GUFREKBKBTBH	
Seller_Order	Book_ID	Seller_Order_Specific_Book_Qty	Return or not	Return Reason	Click to Add			
SOI0015	Database System Concepts	30	☐					
SOI0016	Old Man and Sea	30	☐					
*		0	☐					
		0		\$0.00				

Figure35: Output Screenshot2

ID2.5: Management Report

2.5.1 Order invoice for the Seller Order and Seller Order Item details

This report is highly valuable to senior management as it provides detailed information about sales orders, including customer details, dates, total quantity, and total sales amount. These data enable senior management to monitor sales activities. Additionally, the invoice report for each order includes specific book details, which help management understand current market demand for books and identify trends.

*Annotation: If you want to view the invoice for the **Seller_Order Report**, first click the **Sales Inquiries and Return** button to enter the form, then double-click the **Order_ID** of the order to open the corresponding report.*

Seller_Order_ID	Stu_ID	Last_Name	First_Name	Order_Date	Order_Qty	Freight_Fee	Order_Price	Book_ID	ISBN	Book_Name	Book_Qty
SO0003	SO003	Zhang	Xinyuan	10/12/2024	50	10	\$2,960.00	B0004	004-4	Algorithms	30
								B0003	003-3	Software Enginee	20

Figure36: Seller_Order Report Form View

Seller_Order_ID	Stu_ID	Last_Name	First_Name	Order_Date	Order_Qty	Freight_Fee	Order_Price	Book_ID	ISBN	Book_Name	Book_Qty
SO0003	SO003	Zhang	Xinyuan	10/12/2024	50	10	\$2,960.00	B0004	004-4	Algorithms	30
								B0003	003-3	Software Enginee	20

Figure37: Seller_Order Report Design View

2.5.2 Order invoice for the Publisher Buyer Order and Publisher Buyer Order Item details

This report reflects the company's procurement of books from publishers. It helps

senior management monitor the cost of purchasing second-hand books and evaluate the profitability of the repurchase program. The report also lists the detailed book items for each purchase order, making it more specific and informative.

*Annotation: If you want to view the invoice for the **Publisher_Buyer_Order**, first click the **Purchase Inquiries and Return** button to enter the form, then double-click the **Order_ID** of the order to open the corresponding report. The same process applies to viewing the invoice for the **Student_Buyer_Order**, which will not be elaborated further here.*

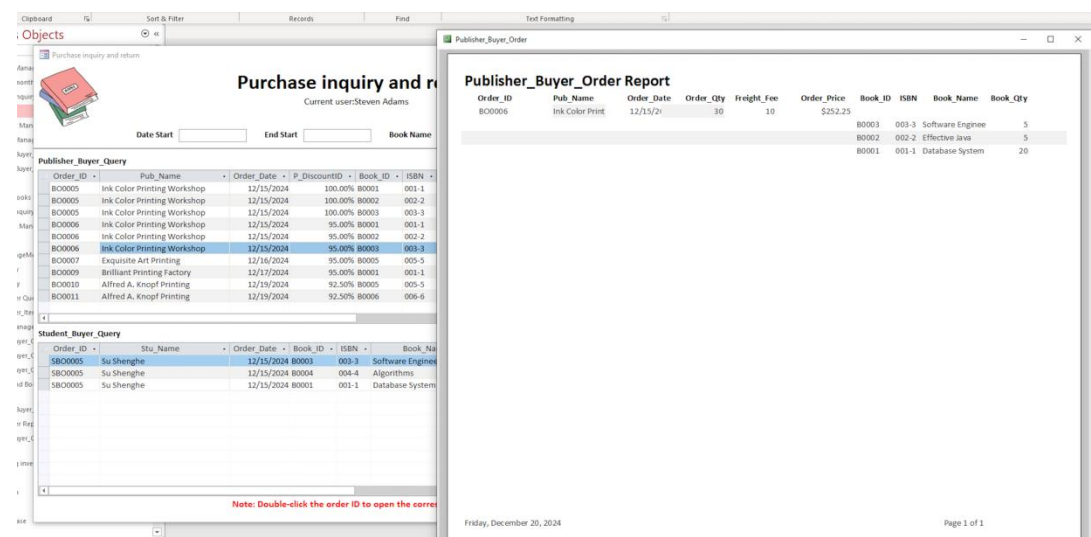


Figure38: Publisher_Buyer_Order Report Form View

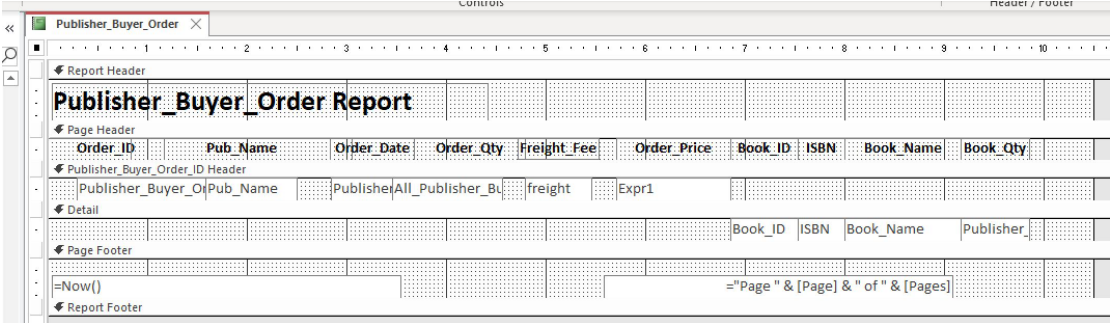


Figure39: Publisher_Buyer_Order Report Design View

ID2.6: Non-trivial chart

2.6.1 Employee Monthly Sales Chart

This bar chart shows book sales over a specific period, with customizable start and end dates. Queries can filter individual employee performance, helping management assess business performance, reward high achievers, and address underperformance.

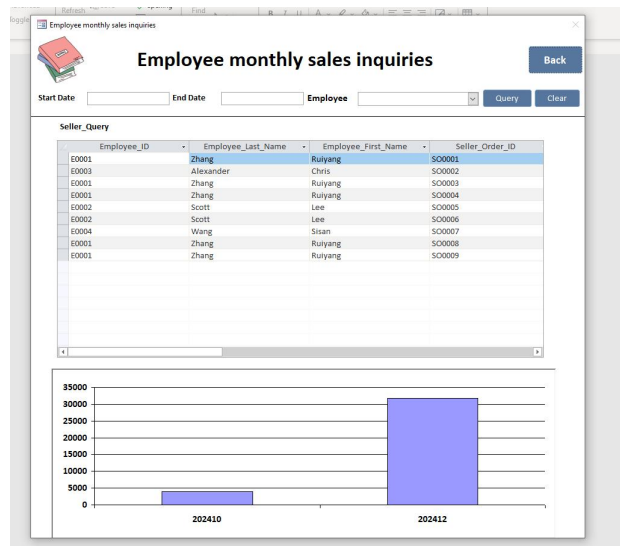


Figure40: Employee Monthly Sales Chart

Next, conduct a detailed search for employees.

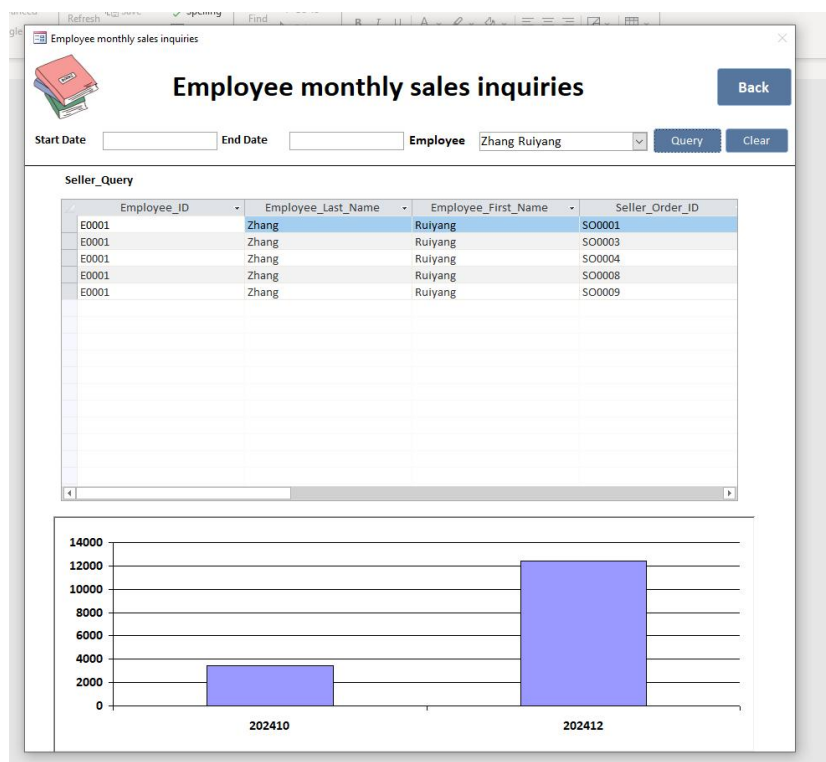


Figure41: Detailed search for employees

Finally, restrict the time period.

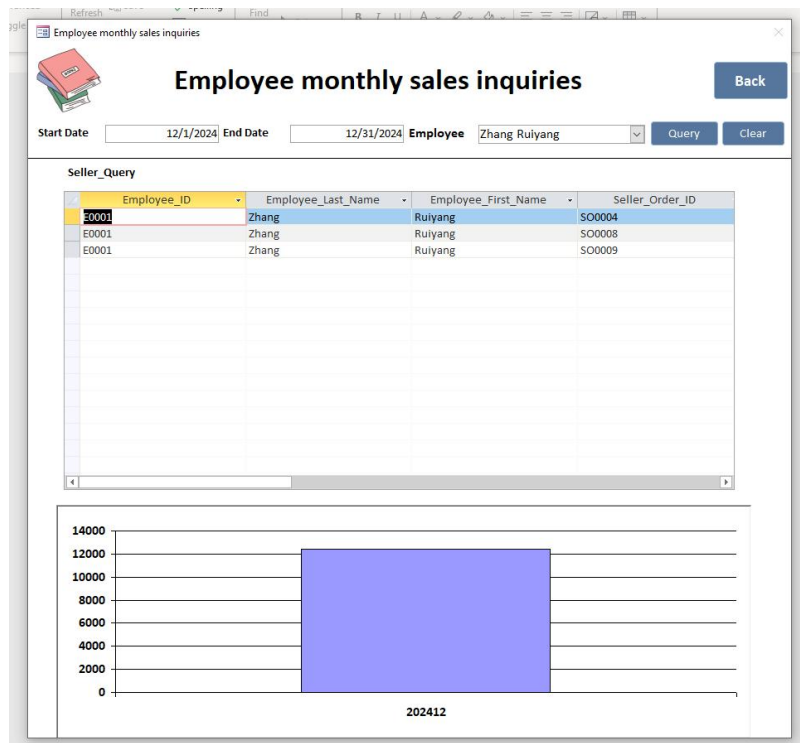


Figure42: Specifically restrict the time period

The Design View is below:

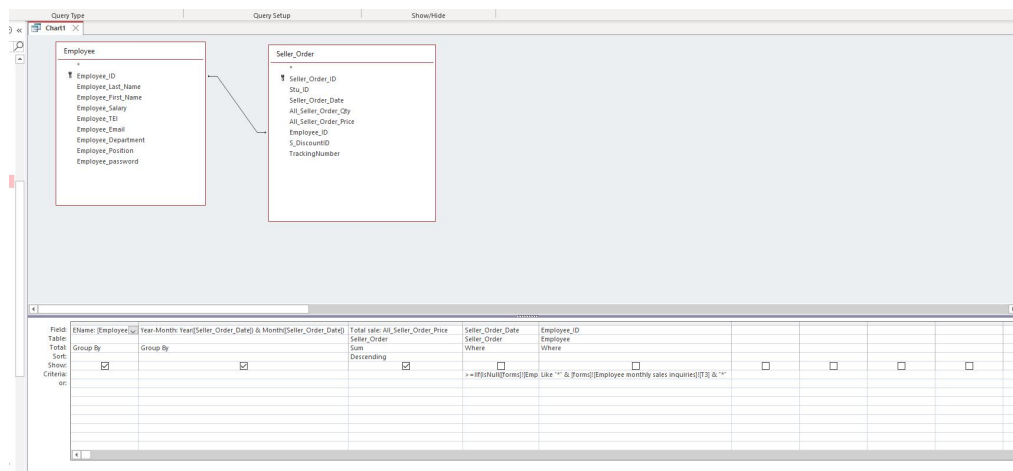


Figure43: The Design View of Figure39

ID2.7: Testing

2.7.1 Add new publisher and books

Original books and suppliers are below:

Inventory inquiry

ISBN Book_Name Pub_Name

Book_ID	ISBN	Book_Name	Current_Qty	Low_Qty	Set_Qty	Author	Marked_Price	Pub_Name
B0004	004-4	Algorithms	152	100	300	Kevin Wayne	\$100.00	Ink Color Printing Workshop
B0005	005-5	Old Man and Sea	50	60	40	Ernest Miller Hemingw	\$160.00	Exquisite Art Printing
B0001	001-1	Database System Concepts	81	80	40	S.Sudarshan	\$20.00	Brilliant Printing Factory
B0002	002-2	Effective Java	141	50	50	Joshua Blosh	\$30.00	Brilliant Printing Factory
B0003	003-3	Software Engineering	126	80	100	Ian Sommerville	\$60.00	Creativity Beauty Printing Bureau

Figure44: Original books and suppliers

Next, we add new publishers and books.

Supplier and Book Management

Query:Pub_Name

Pub_ID Pub_Name

Pub_Address Publisher_TEL

Book

Book_ID	ISBN	Book_Name	Current_Qty	Low_Qty	Set_Qty	Author	Marked_Price
B0006	006-6	Love in the Time of Cholera	0	55	30	Gabriel Garcia Ma	60
*			0	0	0		\$0.00

Figure45: Add new publisher and book

The result, as shown in the figure, is accurate and functions properly.

Book x Publisher x

Book_ID	ISBN	Book_Name	Current_Qty	Low_Qty	Set_Qty	Author	Marked_Price	Pub_ID	Click to Add
B0001	001-1	Database System Concepts	81	80	40	S.Sudarshan	\$20.00	P0003	
B0002	002-2	Effective Java	141	50	50	Joshua Blosh	\$30.00	P0003	
B0003	003-3	Software Engineering	126	80	100	Ian Sommerville	\$60.00	P0004	
B0004	004-4	Algorithms	152	100	300	Kevin Wayne	\$100.00	P0001	
B0005	005-5	Old Man and Sea	50	60	40	Ernest Miller Hemingwa	\$160.00	P0002	
B0006	006-6	Love in the Time of Cholera	0	55	30	Gabriel Garcia Maquez	\$60.00	P0005	
*			0	0	0		\$0.00		

Inventory inquiry

ISBN Book_Name Pub_Name

Book_ID	ISBN	Book_Name	Current_Qty	Low_Qty	Set_Qty	Author	Marked_Price	Pub_Name
B0004	004-4	Algorithms	152	100	300	Kevin Wayne	\$100.00	Ink Color Printing Workshop
B0005	005-5	Old Man and Sea	50	60	40	Ernest Miller Hemingw	\$160.00	Exquisite Art Printing
B0001	001-1	Database System Concepts	81	80	40	S.Sudarshan	\$20.00	Brilliant Printing Factory
B0002	002-2	Effective Java	141	50	50	Joshua Blosh	\$30.00	Brilliant Printing Factory
B0003	003-3	Software Engineering	126	80	100	Ian Sommerville	\$60.00	Creativity Beauty Printing Bureau
B0006	006-6	Love in the Time of Cholera	0	55	30	Gabriel Garcia Maquez	\$60.00	Alfred A. Knopf Printing

Figure46: The output result of the books

Book x Publisher x

Pub_ID	Pub_Address	Pub_Name	Publisher_TEL	Click to Add
P0001	No. 123, Sunshine Street, Garden District, City A	Ink Color Printing Workshop	67885645	
P0002	Room 405, Building B, Fortune Plaza, Central District, City B	Exquisite Art Printing	18879636780	
P0003	Flat 808, Star Tower, Riverside District, City C	Brilliant Printing Factory	63224560	
P0004	No. 567, Green Road, High-tech Zone, City D	Creativity Beauty Printing Bureau	77688914	
P0005	No. 2101, British Cambridge Street, Grass District, City E	Alfred A. Knopf Printing	3847928650	

Inventory inquiry

ISBN Book_Name Pub_Name

Book_ID	ISBN	Book_Name	Current_Qty	Low_Qty	Set_Qty	Author	Marked_Price	Pub_Name
B0004	004-4	Algorithms	152	100	300	Kevin Wayne	\$100.00	Ink Color Printing Workshop
B0005	005-5	Old Man and Sea	50	60	40	Ernest Miller Hemingw	\$160.00	Exquisite Art Printing
B0001	001-1	Database System Concepts	81	80	40	S.Sudarshan	\$20.00	Brilliant Printing Factory
B0002	002-2	Effective Java	141	50	50	Joshua Blosh	\$30.00	Brilliant Printing Factory
B0003	003-3	Software Engineering	126	80	100	Ian Sommerville	\$60.00	Creativity Beauty Printing Bureau
B0006	006-6	Love in the Time of Cholera	0	55	30	Gabriel Garcia Maquez	\$60.00	Alfred A. Knopf Printing

Figure47: The output result of the publishers

2.7.2 The functionality of purchasing and selling books, along with the line chart statistics of book sales

At first, *Love in the Time of Cholera* did not display any sales.

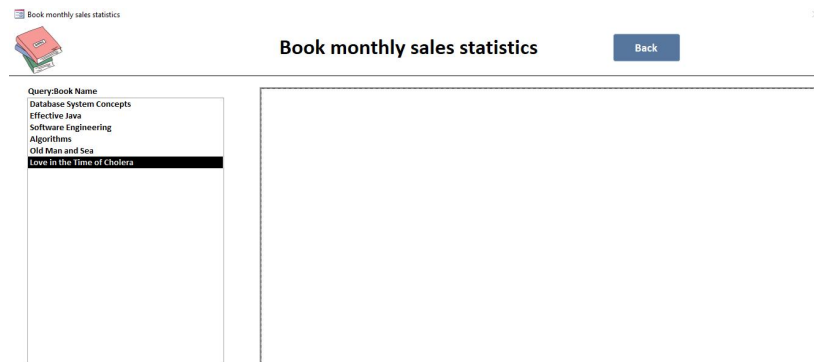


Figure48: The initial chart of *Love in the Time of Cholera*

The initial purchase orders and inventory levels.

Publisher_Buyer_Order_ID	Pub_ID	Publisher_Buyer_Order_Date	All_Publisher_Buyer_Order_Qty	All_Publisher_Buyer_Order_Price	P_Discount%	Employee_ID	TrackingNumber
B00001	P0003	10/3/2024	110	\$864.00		E0003	TRK1234567890
B00002	P0004	10/2/2024	15	\$270.00		E0002	TRK0987654321
B00003	P0001	10/5/2024	60	\$1,665.00		E0004	TRK1122334455
B00004	P0003	10/10/2024	250	\$5,512.50		E0002	TRK5566778899
B00005	P0001	12/15/2024	22	\$147.00	100.00%	E0005	123
B00006	P0001	12/15/2024	30	\$242.25	95.00%	E0005	12312
B00007	P0002	12/16/2024	40	\$1,824.00	95.00%	E0005	HUOFWUBKJQVVB
B00008	P0002	12/16/2024	40	\$1,824.00	95.00%	E0006	DHUGF8HKBQVH
B00009	P0003	12/17/2024	30	\$171.00	95.00%	E0005	DSNFIARBNFSDH
B00010	P0005	12/19/2024	60	\$2,664.00	92.50%	E0005	AJHFOWDBGRTH
			0	\$0.00			

Figure49: The initial purchase orders

Book_ID	ISBN	Book_Name	Current_Qty	Low_Qty	Set_Qty	Author	Marked_Price	Pub_Name
B0004	004-4	Algorithms	152	100	300	Kevin Wayne	\$100.00	Ink Color Printing Workshop
B0005	005-5	Old Man and Sea	110	60	40	Ernest Miller Hemingw	\$160.00	Exquisite Art Printing
B0001	001-1	Database System Concepts	81	80	40	S.Sudarshan	\$20.00	Brilliant Printing Factory
B0002	002-2	Effective Java	141	50	50	Joshua Blosh	\$30.00	Brilliant Printing Factory
B0003	003-3	Software Engineering	126	80	100	Ian Sommerville	\$60.00	Creativity Beauty Printing Bureau
B0006	006-6	Love in the Time of Choler	0	55	30	Gabriel Garcia Marquez	\$60.00	Alfred A. Knopf Printing

Figure50: The initial inventory levels

Next, we purchased **60 copies** of *Love in the Time of Cholera*.

Purchase Books from Publisher

Purchase books from Publisher

Publisher_Buyer_Order_ID: Buyer_Order_City:
 TrackingNumber: P_DiscountID:
 Publisher: Discount_Order_Price:
 PBuyer_Order_Date: Employee:
 Publisher_Buyer_Order_Item:

Book_ID	Book_Qty	Price
Love in the Time of Cholera	60	\$18.00
*	1	

Figure51: Purchase 60 copies of Love in the Time of Cholera

The output result is correct; the order was successfully added, and the inventory was successfully updated. As the **Current_Qty** exceeded the **Low_Qty**, the red highlight on the book was removed.

Publisher_Buyer_Order_ID	Pub_ID	Publisher_Buyer_Order_Date	All_Publisher_Buyer_Order_Qty	All_Publisher_Buyer_Order_Price	P_DiscountID	Employee_ID	TrackingNumber
B00001	P0003	10/3/2024	110	\$864.00	E0003		TRK1234567890
B00002	P0004	10/2/2024	15	\$270.00	E0002		TRK0987654321
B00003	P0001	10/5/2024	60	\$1,665.00	E0004		TRK1122334455
B00004	P0003	10/10/2024	250	\$5,512.50	E0002		TRK5566778899
B00005	P0001	12/15/2024	22	\$147.00	100.00% E0005		123
B00006	P0001	12/15/2024	30	\$242.25	95.00% E0005		12312
B00007	P0002	12/16/2024	40	\$1,824.00	95.00% E0005		HUOFAWUBKIDVB
B00008	P0002	12/16/2024	40	\$1,824.00	95.00% E0006		DHUIGRHKBDVH
B00009	P0003	12/17/2024	30	\$171.00	95.00% E0005		DSNFIARBNFSD
B00010	P0005	12/19/2024	60	\$2,664.00	92.50% E0005		AJHFOWDGBGRTH
B00011	P0005	12/19/2024	60	\$999.00	92.50% E0005		CBCUIREGOSRTNLGS

Publisher_Buyer_Order_Item_ID	Book_ID	Publisher_Buyer_Order_Specific_Book_Qty	Return or nc	Return Reas	Click to Add
B0I0022	B0006	60	☐		
*		0	☐		

Figure52: The order was add successfully

Inventory inquiry

ISBN: Book_Name: Pub_Name:

Book_ID	ISBN	Book_Name	Current_Qty	Low_Qty	Set_Qty	Author	Marked_Price	Pub_Name
B0004	004-4	Algorithms	152	100	300	Kevin Wayne	\$100.00	Ink Color Printing Workshop
B0005	005-5	Old Man and Sea	110	60	40	Ernest Miller Hemingw	\$160.00	Exquisite Art Printing
B0001	001-1	Database System Concepts	81	80	40	S.Sudarshan	\$20.00	Brilliant Printing Factory
B0002	002-2	Effective Java	141	50	50	Joshua Blosh	\$30.00	Brilliant Printing Factory
B0003	003-3	Software Engineering	126	80	100	Ian Sommerville	\$60.00	Creativity Beauty Printing Bureau
B0006	006-6	Love in the Time of Choler	60	55	30	Gabriel Garcia Marquez	\$60.00	Alfred A. Knopf Printing

Figure53: The inventory was updated successfully

Next, we sold **30 copies**.

Sales Management

Order_ID: Order_Qty:

Student: S_DiscountID:

Order_Date: Order_Price:

Employee: TrackingNumber:

Seller_Order_Item

Book	Book_Qty	Price
Love in the Time of Cholera	30	\$48.00
*	1	

Figure54: Sell 30 copies of Love in the Time of Cholera

The order was successfully created. Additionally, as the **Current_Qty** fell below the **Low_Qty**, the book was highlighted in red.

Publisher_Buyer_Order X Student_Buyer_Order X Seller_Order_Item X Seller_Order X

Seller_Order_ID	Stu_ID	Seller_Order_Date	All_Seller_Order_Qty	All_Seller_Order_Price	Employee_ID	S_DiscountID	TrackingNumber	Click to Add
SO0001	S0001	10/4/2024	10	\$466.00	E0001		ZX123456789CN	
SO0002	S0002	10/8/2024	30	\$586.00	E0003		MN987654321CN	
SO0003	S0003	10/12/2024	50	\$2,950.00	E0001		PQ456789123CN	
SO0004	S0003	12/16/2024	60	\$4,725.00	E0001	87.50%	DBUAGFJSBOCV	
SO0005	S0001	12/16/2024	80	\$3,071.25	E0002	87.50%	DBQWCUDSKBREG	
SO0006	S0001	12/16/2024	124	\$8,308.13	E0002	87.50%	HGUFAUBKDBVUG	
SO0007	S0002	12/16/2024	110	\$8,032.50	E0004	87.50%	FWAGDFSBTHGNB	
SO0008	S0004	12/16/2024	30	\$3,888.00	E0001	90.00%	HUIEABWFIUBGGT	
SO0009	S0001	12/17/2024	60	\$3,780.00	E0001	87.50%	GUFREKKBIBTH	
SO0010	S0003	12/19/2024	30	\$1,296.00	E0003	90.00%	BCUIABUGRG	

Seller_Order: Book_ID: Seller_Order_Specific_Book_Qty: Return or not: ☐ Return Reas: Click to Add:

0 0 \$0.00

Sales Inquiry

Current user: Alexander Chris

Start Date: End Date: Book Name:

Sales_Query

Order_ID	Stu_Name	Order_Date	P_DiscountID	Book_ID	ISBN	Book_Name	Book_Qty	Return or not
SO0010	Zhang Xinyuan	12/19/2024	90.00%	B0006	006-6	Love in the Time of Cholera	30	☐
SO0002	Su Shenghe	10/8/2024		B0001	001-1	Database System Concepts	10	☐
SO0002	Su Shenghe	10/8/2024		B0002	002-2	Effective Java	20	☐

Figure55: The order was created successfully

Inventory inquiry

ISBN: Book_Name: Pub_Name:

Book_ID	ISBN	Book_Name	Current_Qty	Low_Qty	Set_Qty	Author	Marked_Price	Pub_Name
B0004	004-4	Algorithms	152	100	300	Kevin Wayne	\$100.00	Ink Color Printing Workshop
B0005	005-5	Old Man and Sea	110	60	40	Ernest Miller Hemingw	\$160.00	Exquisite Art Printing
B0001	001-1	Database System Concepts	81	80	40	S.Sudarshan	\$20.00	Brilliant Printing Factory
B0002	002-2	Effective Java	141	50	50	Joshua Blosh	\$30.00	Brilliant Printing Factory
B0003	003-3	Software Engineering	126	80	100	Ian Sommerville	\$60.00	Creativity Beauty Printing Bureau
B0006	006-6	Love in the Time of Cholera	30	55	30	Gabriel Garcia Marquez	\$60.00	Alfred A. Knopf Printing

Figure56: The inventory was updated successfully

Finally, our line chart dynamically displayed the sales for December 2024 for the book, showing **30 copies** sold. The functionality was successfully implemented.

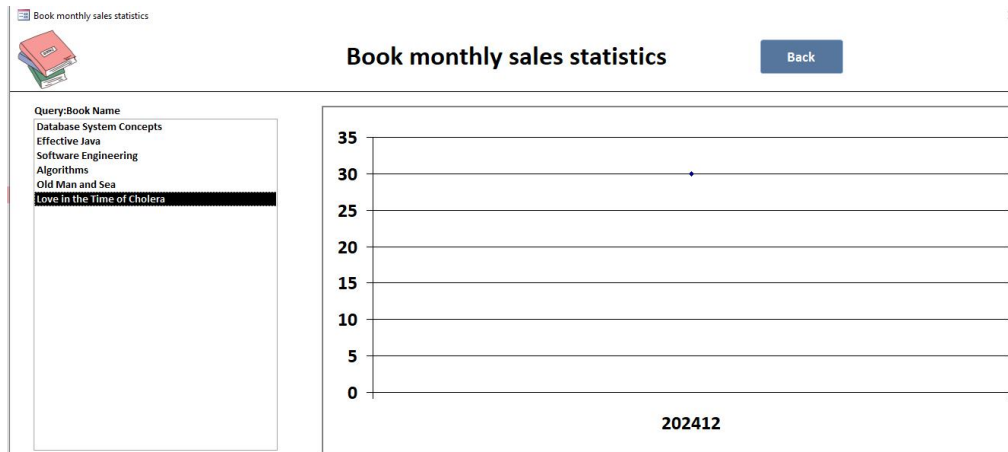


Figure57: The monthly sales chart of Love in the Time of Cholera

ID2.8: Reflection

2.8.1 Advantages

The system employs decentralized management to clarify departmental responsibilities and improve efficiency. Discounts were categorized into a separate table for easier price calculations, and automated discount and price calculations enhance accuracy and reduce errors. Order processing integrates inventory updates and order creation, minimizing employee workload. Monthly sales charts reveal trends for book procurement, while employee sales data encourage performance. Books with **Current_Qty** below **Low_Qty** are highlighted in red, prompting timely restocking.

2.8.2 Disadvantages

Limitations of Returns

First, the limitation of returns requires customers to return the entire quantity of books from a single order detail, which is highly inconvenient. This rigid approach does not allow partial returns, creating dissatisfaction for customers who may only wish to return a portion of the books, thereby reducing the system's flexibility.

Lacks password recovery

Second, the lack of a "Forgot Password" or password recovery feature makes it impossible for employees to regain access if they forget their passwords. This results in operational inefficiency and a potential security risk due to reliance on manual password reset processes.

2.8.3 Solutions and self-evaluation

Flexible Partial Return Process

Introducing a flexible partial return process by adding a field for specifying the return quantity will allow customers to return only the books they need to, significantly improving user experience and operational flexibility.

Secure Password Recovery System

Implementing a secure password recovery system with a “Forgot Password” feature will enable employees to recover their passwords through a security question or email-based reset. Integrating multi-factor authentication (MFA) with external systems will further strengthen system security by preventing unauthorized access (Ciotirnae and Anton, 2024).

Self-evaluation

Familiarizing myself with the system deepened my understanding of database operations, improved my data analysis and problem-solving skills, and enhanced my attention to detail, strengthening both my technical and analytical abilities.

(Word Count: 1625)

References

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<https://search.ebscohost.com/login.aspx?direct=true&db=cat01010a&AN=xjtlu.0002353960&site=eds-live&scope=site> (Accessed: 19 December 2024).